System Requirements Document (SRD)

JRB CHL Enforcement Application

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Change Control

The contents of this document are subject to change control

Application Name: JRB CHL Enforcement
Platform: Android and iOS
Primary Users: Authorized Enforcement Agents, Principal Officers, Supervisors, System Administrators
Integration: Existing JRB CHL Backend and Authentication Infrastructure

1. Purpose

The JRB CHL Enforcement Mobile Application shall provide authorized field enforcement agents with a mobile platform to verify haulage vehicles, validate JRB E-Tags, determine the status of Consolidated Haulage Levy (CHL) trips, take enforcement actions, capture evidence, generate violations and payment demands, and synchronize enforcement activity with the central JRB CHL platform.

The application shall leverage the existing JRB CHL authentication, vendor, agent, vehicle, RFID, QR code, trip, payment and transaction infrastructure through mobile-specific API endpoints.

2. System Objectives

The application shall enable an authorized agent to:

1. Log in using existing JRB CHL credentials.
2. Verify a vehicle through:
    - RFID E-Tag
    - QR Code
    - Manual plate number entry
    - OCR-based vehicle plate scanning
3. Determine whether a vehicle has a valid JRB E-Tag.
4. Determine whether the vehicle has an active CHL trip.
5. View current trip information and validity status.
6. View the most recent CHL trip where no active trip exists.
7. Identify enforcement breaches.
8. Issue violations.

9. Capture enforcement evidence.
10. Generate demand notices and payment requests.
11. Monitor live payment confirmation.
12. Activate the appropriate post-payment process, including creation of a CHL trip where applicable.
13. Capture mandatory GPS and enforcement location data.
14. Operate with limited connectivity using offline caching and synchronization.
15. Maintain a complete audit trail of all verification and enforcement activities.
16. Receive notifications from backend.
17. Send notifications to backend
    - SOS
    - Report Vehicle / Driver / Situation
    - Notify system
    - Confirm loaded / empty status

3. Scope

3.1 In Scope

Authentication

- Existing JRB CHL authentication
- Email and password login
- Forgot password
- Token/session management
- Role-based access
- Automatic logout after 24 hours
- Immediate access denial when an account is deactivated centrally

Vehicle Verification

- UHF RFID reading

- QR code scanning
- Manual plate number verification
- OCR vehicle plate recognition
- Active trip lookup
- E-Tag validation
- Vehicle identification
- Most recent trip lookup

Enforcement

- Violation issuance
- Evidence capture
- GPS capture
- Notes
- Demand notice generation
- Live payment monitoring
- Violation lifecycle management

Offline Capability

- Local data cache
- Offline verification where cached data is available
- Offline evidence captures
- Offline violation creation
- Automatic synchronization when connectivity returns

Monitoring

- Command and Control Centre Dashboard
- Monitoring and Corporate Dashboard
- Agent dashboard
- Supervisor dashboard
- Verification history

- Audit trail
- Repeat vehicle reporting alerts

Integrations

Integration with the following systems

- CHL
- EDMS
- Enforcement Engine
- CCA
- MVLAs
- Other Systems

3.2 Out of Scope

The application shall not independently manage:

- Vendor creation
- Principal Officer creation
- Agent account creation
- Core vehicle registration
- Core RFID tag issuance and registration
- Core CHL tariff configuration
- Core trip creation rules

These functions already exist within the central JRB CHL platform.

4. System Users and Roles

4.1 Enforcement Agent

The Enforcement Agent shall be able to:

- Log in
- Verify vehicles

- Use RFID, QR, plate entry and OCR
- View vehicle and trip status
- Issue violations
- Capture evidence
- Add enforcement notes
- Generate demand notices
- View live payment status
- View personal verification history
- View recent activity
- Perform offline actions

Agents shall not be restricted from verifying a vehicle based solely on their assigned geographical scope.

Their enforcement scope may, however, be defined within the central system as:

- Enforcement Location
- Enforcement Zone
- Ward
- Local Government Area
- State
- Region
- Subregion
- National

4.3 Supervisor

The Supervisor shall be able to monitor:

- Agent verification activity
- Active and inactive trip results
- Violations

- Unresolved cases
- Agent activity by location
- Repeat vehicles reported without appropriate follow-up
- Enforcement performance

4.4 System Administrator

The System Administrator shall have access to the complete audit trail and central system administration functions.

The System Administrator may also close or cancel violations where authorized.

5. Authentication and Session Requirements

5.1 Login

The application shall provide:

- Email field
- Password field
- Login button
- Forgot Password link

Authentication shall use the existing JRB CHL authentication service.

5.2 Forgot Password

The application shall integrate with the existing password recovery process.

The exact reset process shall be controlled by the central authentication system.

5.3 Session Management

The application shall:

- Maintain a secure authenticated session.
- Use access and refresh tokens.

- Automatically expire the session after 24 hours.
- Require the agent to log in again after expiry.
- Deny access if the agent account has been deactivated.
- Log the user out when central authentication confirms that the account is no longer active.

5.4 Security Requirements

The mobile API architecture shall support:

- HTTPS
- Token-based authentication
- Access tokens
- Refresh tokens
- Secure token expiry
- Role-based authorization

6. Application Navigation

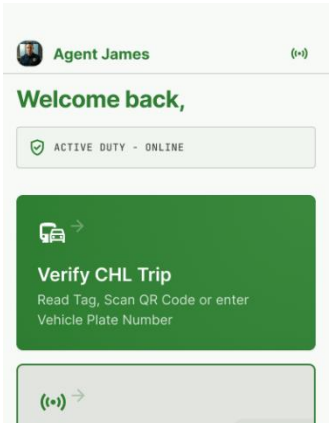
The primary application structure shall include:

Main Navigation

1. Dashboard
2. Verification History
3. Notifications
4. Profile/Settings

The exact navigation layout may be implemented using a bottom navigation structure, or another field-friendly Android or iOS navigation pattern.

The design must prioritize rapid access to verification.



7. Dashboard Requirements

7.1 Agent Dashboard

The dashboard shall display the following daily metrics:

- Verify CHL Trip
- Verify E-Tag
- Total Verifications
- Violations Issued

Current Operational Status

The dashboard shall also show:

- RFID Reader Status
- Internet Connection Status
- GPS Status
- Offline Synchronization Status

Recent Activity

The dashboard shall display the agent's most recent five verifications.

Each recent activity item should include, where available:

- Vehicle plate number
- Verification method
- Verification result
- Time
- Action status

The agent shall be able to access a full activity history.

7.2 Supervisor Dashboard

The supervisor view shall include:

- Total Agent Verifications
- Active Trip Results
- Inactive/No Active Trip Results
- Violations by Agent

- Unresolved Violations
- Repeat Vehicles Without Action
- Agent Activity by Location

8. Verify CHL Trip

The Verify CHL Trip function is the primary operational workflow of the application.

The agent shall be presented with four verification methods:

1. Read Haulage Tag
2. Scan QR Code
3. Enter Vehicle Plate Number
4. Scan Vehicle Plate

All verification methods shall ultimately resolve to the same backend verification service.

Logical Architecture

Verification Method → Vehicle Identification → Vehicle Record → E-Tag Validation → Active CHL Trip Lookup → Verification Result

9. Haulage Tag Verification

9.1 RFID Standard

The application shall support **UHF RFID** technology.

9.2 Supported Reader Types

The application shall support:

1. USB-C RFID readers
2. Fixed RFID readers
3. Built-in RFID readers
4. Handheld RFID sled devices

9.3 Reader Activation

The Mobile application shall not continuously scan for RFID readers.

RFID hardware detection and communication shall be activated only when the agent selects the RFID verification method.

This requirement is intended to reduce unnecessary battery consumption.

9.4 Mobile Reader Flow

The standard flow shall be:

Agent selects Read Haulage Tag
→ Application activates reader
→ Reader obtains Tag ID and Reference Number
→ Application submits Reference Number to Backend
→ Backend identifies Vehicle
→ Backend validates E-Tag
→ Backend checks CHL Trip Status
→ Application displays Result

Raw RFID values shall never be displayed to the enforcement agent.

9.5 Fixed Reader Architecture

Fixed readers shall communicate through:

Fixed RFID Reader → WebSocket Gateway/Service → Backend → Mobile Application

The Mobile application shall receive the processed or relevant reader event through the approved backend architecture.

The mobile application shall not require a direct WebSocket connection to a fixed reader unless explicitly introduced in a future technical design.

9.6 Multiple Tag Detection

Where multiple RFID tags are detected:

1. The application shall submit detected tag references for backend resolution.
2. The backend shall identify the associated vehicles.

3. The application shall display the identified vehicle plate numbers.
4. The agent shall select the vehicle to be verified.

The raw RFID values shall remain hidden.

9.7 RFID Error States

Reader Disconnected

Display:

RFID Reader Disconnected

No Tag Detected

The application may:

- Continue scanning, or
- Display **No Tag Detected**

The system shall not automatically classify the vehicle as non-compliant simply because a reader scan has not detected a tag.

Unknown Tag

Display:

Tag Not Part of JRB CHL Architecture

This shall be treated as an enforcement breach.

Repeated Scan

Repeated scans of the same tag shall not generate repeated backend requests unnecessarily.

The application shall cache the immediately processed tag and suppress duplicate requests for the same scan event.

10. QR Code Verification

10.1 QR Code Structure

The vehicle's physical JRB E-Tag shall contain a QR code.

The QR code shall represent the reference number associated with the vehicle.

The RFID Tag ID, Reference Code and QR Code reference shall all resolve to the appropriate vehicle record within the JRB CHL architecture.

10.2 QR Verification Flow

Agent selects Scan QR Code
→ Mobile Device Camera Opens
→ QR Code is Detected
→ Application Extracts Reference Number
→ Reference Number is Sent to Backend
→ Backend Resolves Vehicle
→ E-Tag and Trip Status are Verified
→ Result is Displayed

The application shall use the Mobile device camera and an integrated QR scanning capability.

11. Manual Plate Number Verification

The application shall provide a vehicle plate number input field.

The verification process shall:

- Ignore spaces
- Ignore hyphens
- Perform case-insensitive matching
- Normalize the plate before submitting the request

Example:

ABC 123 AA
ABC-123-AA
abc123aa

shall be normalized to the same search value where applicable.

12. OCR Vehicle Plate Verification

OCR plate recognition shall be included in Version 1.

12.1 OCR Flow

Agent selects Scan Vehicle Plate
→ Camera Opens
→ Plate is Detected
→ OCR Extracts Plate Number
→ Application Normalizes Result
→ Agent Confirms or Edits Plate
→ Plate is Submitted for Verification

12.2 OCR Processing

OCR shall operate:

- On-device where possible.
- With online or alternative fallback where supported.
- With mandatory agent confirmation before verification.

The OCR result shall never be automatically treated as final without allowing the agent to correct it.

13. Vehicle and E-Tag Verification Logic

The application shall distinguish between vehicle identity, E-Tag validity and CHL trip status.

13.1 E-Tag Missing

Where an agent physically inspects a vehicle and confirms that no JRB E-Tag is present:

Result:

E-Tag Missing — Violation

Available actions shall include:

- Issue Violation
- Capture Evidence
- Generate demand notice
- Apply applicable violation fee
- Trigger the process for tagging the vehicle after payment

13.2 Invalid or Fake E-Tag

Where an RFID tag cannot be resolved to the JRB CHL architecture:

Result:

Invalid E-Tag — Not Part of JRB CHL Architecture

This shall automatically qualify as an enforcement violation.

13.3 Tag-Vehicle Mismatch

Where the RFID or QR record resolves to one vehicle but the physical vehicle has a different plate:

Result:

Tag-Vehicle Mismatch — Violation

The application shall require or strongly enforce:

- Physical vehicle plate capture
- Vehicle photograph
- GPS
- Enforcement notes

13.4 Valid E-Tag with No Active CHL Trip

Where:

- The E-Tag is valid, and
- The vehicle is identified, but
- No active CHL trip exists

The application shall display:

Valid E-Tag — No Active CHL Trip

Enforcement actions shall be presented only where the vehicle is confirmed to be carrying haulage goods.

The agent shall determine whether the vehicle is actively engaged in haulage before initiating the applicable CHL enforcement process.

14. Active CHL Trip Definition

A vehicle shall have a maximum of one active CHL trip at a time.

An active trip shall be determined through backend business rules.

Trip validity shall consider:

- Payment status
- Origin
- Destination
- Distance
- Expected travel duration
- Configured backend rules
- Destination arrival detection
- Vehicle scan activity

The backend shall determine the authoritative trip status.

14.1 Trip Expiry

Trip duration shall be determined using a combination of:

- Distance
- Origin and destination
- Backend configuration

A trip that exceeds its expected travel duration shall not automatically be treated as invalid solely because of elapsed time.

Vehicles may experience:

- Traffic delays
- Mechanical breakdown
- Other legitimate travel delays

Therefore, trip completion and validity must also consider destination detection.

14.2 Destination Detection

A trip may be identified as having reached its destination when the vehicle is detected within the destination LGA.

The backend shall validate:

- Agent GPS
- Registered enforcement point
- LGA geofence

Trip completion shall not necessarily occur immediately.

The backend may transition the trip to completion within a defined number of hours after valid destination detection, according to central CHL business rules.

15. Trip Verification Results

The application shall support the following result states.

Result	Meaning	Primary Action
Active Trip	Vehicle has a valid E-Tag and an active CHL trip	View trip details
No Active Trip	Vehicle identified but no valid current CHL trip	Enforce where applicable
E-Tag Missing	No JRB E-Tag present	Issue violation
Invalid E-Tag	Tag is not part of JRB CHL architecture	Issue violation
Tag-Vehicle Mismatch	Tag belongs to another vehicle	Issue violation
Vehicle Not Found	Plate cannot be resolved	Record/investigate
No Trip History	Vehicle exists but has never made a CHL trip	Enforce where applicable

Result	Meaning	Primary Action
Verification Unavailable	Live verification cannot be completed	Retry or use offline cache
Expired/Previous Trip	Previous trip exists but is no longer current	View/enforce as applicable

16. Active Trip Result Page

When an active trip is found, the result page shall display:

1. Transaction Reference
2. Trip Start Date/Time
3. Payment Date/Time
4. Vehicle Plate Number
5. Vehicle Type
6. Transaction Type
7. Originating State
8. Originating Local Government
9. Destination State
10. Destination Local Government
11. Trip Status
12. Trip Validity Countdown

The validity countdown should provide a clear operational indicator such as:

Valid for 06h 42m

Where applicable, the backend may return additional trip metadata.

17. No Active Trip Result

Where no active trip is found, the application shall display:

No Active Trip Found

The application shall provide:

View Last CHL Trip

The last trip view shall show the same trip detail structure as an active trip, including its current historical status.

Only the most recent CHL trip shall be displayed.

17.1 No Trip History

Where the vehicle has never made a CHL trip, display:

No Trip History

The system shall not attempt to fabricate historical information.

18. Enforcement Actions

Where enforcement is required, the agent shall be able to:

1. Issue Violation
2. Generate Demand Notice
3. Direct Vehicle/Driver to Pay
4. Generate a payment method or reference
5. Capture evidence
6. Add notes
7. Generate a demand notice
8. Monitor payment confirmation

19. Violation Categories

The initial violation categories shall include:

1. No Active CHL Trip
2. No E-Tag
3. Fake/Invalid E-Tag

4. Tampered E-Tag
5. Expired Trip
6. Refusal to Comply

Violation categories and applicable charges should be retrieved from the backend to allow central configuration.

20. Violation Creation Workflow

The violation workflow shall be:

Verify Vehicle
→ Identify Breach
→ Issue Violation
→ Add Notes
→ Capture Vehicle Photograph(s)
→ GPS Automatically Captured
→ Date/Time Captured
→ Agent Identity Captured
→ Submit Violation
→ Generate Case/Violation Reference
→ Generate Demand Notice/Payment Invoice

The application shall not allow a mandatory evidence requirement to be silently bypassed.

21. Violation Statuses

A violation shall support the following statuses:

- Payment Pending
- Resolved
- Escalated
- Cancelled

Additional internal states may be implemented by the backend where required.

22. Violation Closure Authority

A violation may be closed or cancelled, subject to authorization, by:

- Enforcing Agent
- Principal Officer
- Supervisor
- System Administrator
- Automatic system process after confirmed payment

The backend shall enforce role-based authorization for each action.

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23. Demand Notice Workflow

The payment workflow shall be:

Agent Issues Violation
→ Agent Selects Generate Payment Request
→ System Retrieves Vehicle Details
→ Backend Calculates Applicable CHL Levy + Penalty
→ Payment Request Generated
→ Driver Receives Payment Option/Reference
→ Payment Confirmed
→ CHL Trip Activated or Relevant Compliance Process Completed

23.1 Supported Payment Methods

The application shall support one or more of:

- QR Code
- USSD
- Bank Transfer
- Generated Bank Account/Reference
- POS/Card Payment
- Existing CHL Payment Platform

The exact available methods shall be returned by the backend or payment service.

23.2 Payment Confirmation

The agent shall be able to see live payment progression where connectivity permits:

Payment Pending → Payment Received → CHL Trip Activated

The system should automatically refresh the relevant enforcement or trip state after payment confirmation.

24. Evidence Capture

24.1 Mandatory Evidence

When a violation is issued, the following shall be mandatory:

- Vehicle photograph
- GPS location
- Date and time
- Agent identity
- Vehicle plate number

24.2 Multiple Photographs

The application shall support multiple photographs, including:

- Front of vehicle
- Rear of vehicle
- Side of vehicle
- Missing E-Tag evidence
- Fake or invalid tag evidence
- Other relevant evidence
- Haulage Bay

24.3 Camera Restriction

Evidence shall be captured directly using the application camera.

Gallery uploads shall not be permitted for enforcement evidence.

24.4 Evidence Metadata

Evidence metadata shall be stored with the record but shall not be visibly watermarked onto the image.

Stored metadata shall include, where available:

- Capture timestamp
- GPS coordinates/location
- Agent identity
- Violation reference

25. GPS and Enforcement Location

GPS capture is mandatory for every verification.

The application shall:

1. Request appropriate location permissions.
2. Capture the agent's location at verification time.
3. Capture the enforcement location where applicable.
4. Associate the location with the verification and enforcement action.

GPS shall support:

- Verification audit trail
- Destination detection validation
- Enforcement point identification
- Violation evidence

26. Offline Operation

The application shall support operation in areas with unreliable connectivity.

The application shall use an offline cache.

26.1 Cached Data

The application shall cache:

- Recently verified vehicles with destinations matching the agent's state
- Active CHL trips
- Violation categories and configuration
- Payment and levy rules

The exact cache synchronization strategy shall be configurable by the backend.

26.2 Offline Verification

If a vehicle is found in the local cache, the application may display the cached verification result.

The interface shall clearly distinguish cached/offline information from a live backend result.

If the vehicle is not available locally, display:

Unable to Verify Offline — Vehicle Not Available in Local Cache

The application shall never incorrectly display No Active Trip merely because a live request could not be completed.

26.3 Offline Action Queue

The application shall queue offline actions, including:

- Verification records
- Violations
- Notes
- Photographs
- GPS records
- Other enforcement activity

Synchronization shall occur automatically when internet connectivity is restored.

The application is not required to display a dashboard count of pending synchronization actions.

27. Verification History

The agent shall have access to:

- The last 30 days of verification history.
- A button to view all available personal history.

Agent history search is not required in Version 1.

Each history item shall be selectable to view its details.

27.1 History Information

Historical records shall indicate:

- Verification completion status
- Action taken
- Violation issued
- Payment generated
- Pending action

28. Audit Trail

Every verification attempt shall be recorded, including successful and unsuccessful attempts.

The audit trail shall include:

- Agent ID
- Agent name
- Vendor/organization where applicable
- Date and time
- Verification method
- Processed identifier
- Vehicle ID where resolved
- Vehicle plate number where resolved
- Verification result
- Trip reference where applicable

- Violation reference where applicable
- GPS location
- Enforcement point
- Device/application metadata where available
- Action taken
- Synchronization status

The System Administrator shall have access to the complete audit trail.

Supervisors shall have access according to role authorization.

29. Repeat Vehicle Alerting

The system shall identify when the same vehicle has been reported multiple times without appropriate follow-up.

The alert threshold shall initially be:

3 reports

After the third qualifying report, the system shall notify the central system/supervisory process so appropriate action can be taken regarding the enforcement workflow and responsible agents.

Alerts shall be delivered through:

- Push notifications
- In-app notifications

30. Application Notifications

The application shall support notifications for:

- RFID reader connected
- RFID reader disconnected
- No internet connection
- Internet restored
- GPS disabled
- Pending offline actions

- Offline actions successfully synchronized
- App update available
- Session expiring
- Repeat vehicle alerts

31. API Requirements

The Mobile application shall consume mobile-specific API endpoints.

The APIs shall integrate with existing JRB CHL systems for:

- Authentication
- Authorization
- Vendor/agent identity
- Vehicle registration
- RFID records
- QR references
- CHL trips
- Payments
- Transactions
- Violations
- Evidence
- Notifications
- Audit trails

31.1 Suggested Mobile API Domains

The API design should logically support:

Authentication

- Login
- Refresh Token

- Logout
- Forgot Password

Dashboard

- Agent dashboard
- Supervisor dashboard

Verification

- Verify by RFID Reference
- Verify by QR Reference
- Verify by Plate Number
- Verify OCR-confirmed Plate
- Retrieve Last CHL Trip

Enforcement

- Create Violation
- Upload Evidence
- Add Notes
- Generate Demand Notice
- Retrieve Payment Status

Synchronization

- Download Offline Cache
- Upload Offline Verification
- Upload Offline Violation
- Upload Evidence
- Resolve synchronization status

History

- Retrieve Agent History
- Retrieve Verification Details

Notifications

- Retrieve Notifications
- Mark Notification Read

The detailed endpoint definitions, request objects and response schemas should be produced as part of the API technical specification.

32. Functional Requirements

FR-01: Login

The system shall authenticate an agent using email and password.

FR-02: Password Recovery

The system shall provide access to the existing forgot-password process.

FR-03: Dashboard

The system shall display operational metrics and device/service status.

FR-04: Haulage Tag Verification

The system shall verify vehicles using supported UHF RFID readers.

FR-05: QR Verification

The system shall verify vehicles through the physical QR code associated with the JRB E-Tag.

FR-06: Manual Plate Verification

The system shall accept normalized plate numbers without requiring spaces or hyphens.

FR-07: OCR Verification

The system shall detect, extract and allow confirmation of vehicle plate numbers.

FR-08: Active Trip Lookup

The system shall retrieve the authoritative current CHL trip status.

FR-09: E-Tag Validation

The system shall identify valid, invalid, missing and mismatched E-Tag scenarios.

FR-10: Last Trip

The system shall display only the most recent CHL trip when no active trip exists.

FR-11: Violation Creation

The system shall allow authorized agents to create violations.

FR-12: Evidence

The system shall require direct camera evidence and automatically collect enforcement metadata.

FR-13: Demand Notice

The system shall generate applicable levy and penalty payment requests.

FR-14: Payment Monitoring

The system shall display payment progression and confirmation.

FR-15: GPS

The system shall capture mandatory GPS information for every verification.

FR-16: Offline Operation

The system shall support cached verification and offline action queuing.

FR-17: Automatic Synchronization

The system shall synchronize queued actions when connectivity is restored.

FR-18: History

The system shall provide personal verification history.

FR-19: Notifications

The system shall deliver operational and enforcement alerts.

FR-20: Audit Trail

The system shall record all verification and enforcement actions.

FR-21: Data Protection

The system shall protect user and system data using global standards as defined by ISO27001.

33. Non-Functional Requirements

33.1 Performance

The application should:

- Open verification functions quickly.

- Prevent duplicate RFID backend calls.
- Display clear loading states during live verification.
- Process cached verification data without requiring network access.

33.2 Reliability

The application shall:

- Preserve offline actions until successful synchronization.
- Avoid duplicate submission of violations.
- Handle interrupted uploads.
- Retry failed synchronization safely.
- Clearly distinguish network failure from a negative verification result.

33.3 Usability

The user interface shall be optimized for field use with:

- Large buttons
- Clear status indicators
- Minimal workflow steps
- High readability
- High-contrast result states
- Clear error messages
- Standard smartphone support
- Rugged device support

33.4 Device Support

The application shall support:

- Standard Android and iOS smartphones
- USB-C RFID reader devices

- Handheld RFID sled devices
- Fixed RFID reader integration points

Recommended minimum Mobile Device version: Android 10 or later, ios 15 or later, subject to final technical confirmation.

33.5 Deployment

The application should support deployment through one or more approved channels:

- Google Play Store
- Direct APK distribution
- Apple Store

The final deployment method shall be determined during release planning.

34. Primary Screen Requirements

Screen 1 — Login

Components:

- JRB CHL branding
- Email
- Password
- Show/hide password
- Login button
- Forgot Password

Screen 2 — Agent Dashboard

Components:

Metrics

- Total Verifications
- Active Trips

- No Active Trips
- Violations Issued

Quick Actions

- Verify CHL Trip

Operational Status

- RFID
- Internet
- GPS

Recent Activity

- Last five verifications

Screen 3 — Select Verification Method

Four primary options:

1. Read RFID E-Tag
2. Scan QR Code
3. Enter Plate Number
4. Scan Vehicle Plate

Screen 4 — RFID Reader

Components:

- Reader status
- Connection status
- Scan state
- Multiple vehicle selection where applicable
- Error messages

The raw RFID Tag ID shall not be displayed.

Screen 5 — QR Scanner

Components:

- Camera preview
- Scan guidance
- QR detection
- Processing state
- Error state

Screen 6 — Manual Plate Entry

Components:

- Plate number field
- Automatic normalization
- Verify button

Screen 7 — OCR Plate Scanner

Components:

- Camera
- Plate detection area
- Captured OCR result
- Edit/confirm field
- Verify button

Screen 8 — Active Trip Result

Components:

- Clear Active status
- Trip details
- Validity countdown
- Origin

- Destination
- Transaction reference

Screen 9 — No Active Trip

Components:

- No Active Trip status
- Vehicle information
- View Last CHL Trip
- Enforcement actions where applicable

Screen 10 — Violation Creation

Components:

- Violation category
- Notes
- Vehicle photographs
- Automatically captured GPS
- Date/time
- Agent identity
- Submit

Screen 11 — Demand Notice

Components:

- Vehicle details
- CHL levy
- Penalty
- Total payable amount
- Available payment methods

- Payment reference
- Payment QR code where applicable
- Live payment status

Screen 12 — Verification History

Components:

- Last 30 days
- View All History
- Verification status
- Action status
- Vehicle details
- Tap to view details

Screen 13 — Notifications

Components:

- Operational notifications
- Sync notifications
- Reader notifications
- Repeat vehicle alerts
- Payment/enforcement notifications

35. Core End-to-End Workflows

Workflow A — Valid Vehicle with Active CHL Trip

Agent Login
→ Dashboard
→ Verify CHL Trip
→ Select Verification Method
→ Identify Vehicle
→ Backend Verification

→ Active Trip
→ Display Trip Details and Validity Countdown
→ Record Verification and GPS
→ Complete

Workflow B — Valid E-Tag but No Active Trip

Verify Vehicle
→ Valid E-Tag Identified
→ No Active CHL Trip
→ Agent Confirms Vehicle is Carrying Haulage
→ Enforcement Options
→ Issue Violation
→ Capture Evidence
→ Generate Demand Notice
→ Payment Confirmed
→ CHL Trip Activated

Workflow C — No E-Tag

Physical Inspection
→ No E-Tag
→ E-Tag Missing — Violation
→ Capture Evidence
→ Issue Violation
→ Generate Demand Notice Including Applicable Fees
→ Payment Confirmed
→ Vehicle Tagging Process Initiated

Workflow D — Invalid E-Tag

RFID Scan
→ Tag Cannot Be Resolved
→ Invalid E-Tag
→ Issue Violation
→ Capture Evidence
→ Generate Applicable Demand/Payment Process

Workflow E — Tag-Vehicle Mismatch

RFID/QR Scan
→ Tag Resolves to Vehicle A
→ Physical Vehicle Plate is Vehicle B
→ Tag-Vehicle Mismatch
→ Capture Plate
→ Capture Vehicle Photographs
→ GPS Captured
→ Issue Violation

Workflow F — Offline Verification

Agent Starts Verification
→ No Internet
→ Search Local Cache
→ Vehicle Found → Display Cached Result
OR
→ Vehicle Not Found → Unable to Verify Offline
→ Record Activity
→ Queue Actions
→ Connectivity Restored
→ Automatic Synchronization

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36. Acceptance Criteria

The application shall be accepted for deployment when the following core criteria are successfully demonstrated:

1. An existing agent can log in using central JRB CHL credentials.
2. A deactivated agent is denied continued access.
3. An agent can verify a vehicle through RFID.
4. An agent can verify a vehicle through QR.
5. An agent can verify a vehicle through manual plate entry.
6. An agent can verify a vehicle through OCR plate scanning.
7. Raw RFID values are not exposed to the agent.
8. Multiple RFID detections allow vehicle selection by plate number.

9. Duplicate RFID scans do not create duplicate backend verification calls.
10. Active trip details are correctly displayed.
11. Trip validity countdown is displayed.
12. No Active Trip is clearly distinguished from an API failure.
13. The most recent CHL trip can be viewed.
14. No Trip History is correctly displayed.
15. E-Tag Missing can trigger enforcement.
16. Invalid E-Tags are identified as not belonging to the JRB CHL architecture.
17. Tag-to-vehicle mismatches can be recorded as violations.
18. GPS is captured for every verification.
19. A violation cannot be submitted without mandatory evidence.
20. Evidence can only be captured through the application camera.
21. Multiple evidence photographs can be captured.
22. A demand notice can include CHL levy and penalty.
23. Live payment confirmation is supported.
24. Appropriate trip/compliance status is updated after payment.
25. The application operates with cached data while offline.
26. Offline actions synchronize automatically.
27. All verification attempts are recorded.
28. Agents can view their history.
29. Supervisors can monitor agent activity.
30. Repeat vehicle reporting is detected after three reports.
31. Repeat alerts are delivered through push and in-app notifications.
32. The complete activity trail is available to authorized administrators.

37. Recommended Technical Architecture

Mobile Application

Android Native Application

Recommended characteristics:

- Modern Android architecture
- Local encrypted data storage
- Background synchronization service
- Camera integration
- GPS/location services
- RFID hardware abstraction layer
- QR scanning module
- OCR module
- Push notification capability

iOS Native Application

- Modern iOS architecture using Swift and SwiftUI
- Local encrypted data storage using Keychain and protected app storage
- Background synchronization using BackgroundTasks
- Native camera integration
- GPS/location services using Core Location
- RFID hardware abstraction layer for supported external readers and handheld sled devices
- Native QR code scanning module
- OCR and vehicle plate recognition module
- Push notification capability using APNs

Integration Layer

JRB CHL Mobile API

Responsible for:

- Authentication

- Authorization
- Vehicle resolution
- Tag validation
- Trip status
- Enforcement
- Payment integration
- Notifications
- Offline synchronization

Central Systems

Existing services for:

- Vendor management
- Principal Officers
- Agents
- Vehicles
- RFID/E-Tag records
- QR references
- CHL trips
- Payments
- Transactions

RFID Architecture

Mobile

RFID Reader / USB-C / Handheld Sled → Mobile App → Mobile API

Fixed Reader

Fixed Reader → WebSocket Gateway/Service → Backend → Mobile API/App

38. Future Extensibility

The architecture should support future addition of:

- Additional violation categories
- New RFID reader models
- Automated vehicle classification
- AI-assisted haulage detection
- Expanded OCR
- Driver identification
- Digital impound workflow
- Supervisor case management
- Advanced geofencing
- Automated fraud detection
- Agent performance scoring
- National enforcement analytics

39. Final Product Definition

JRB CHL Enforcement is a secure Mobile field enforcement application that enables authorized agents to identify haulage vehicles using RFID, QR codes, manual plate entry, or OCR; validate their participation in the JRB CHL architecture; determine their live Consolidated Haulage Levy trip status; and take immediate, auditable enforcement action where a breach is identified.

The application shall function as the field enforcement layer of the wider JRB CHL ecosystem, leveraging the existing backend while adding mobile-specific capabilities for:

Verification → Validation → Location Capture → Evidence → Violation → Demand Notice → Payment → Compliance/Trip Activation → Audit Trail

This SRD is grounded in the requirements and enforcement rules provided for the application, including the defined violation, payment, E-Tag, offline, GPS, RFID, OCR, dashboard, notification, and deployment requirements.